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Multi-objective optimization of multi-energy complementary systems integrated biomass-solar-wind energy utilization in rural areas

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A B S T R A C T

Rural areas possess abundant renewable energy sources, such as solar and biomass energy; however, the current methods of energy utilization suffer from low efficiency and serious pollution issues. As rural residents' living standards continue to improve, there is an urgent need to optimize and adjust the structure of rural energy systems. Multi-energy complementary systems (MECS) have the potential to enhance energy utilization efficiency, achieve high efficiency and energy savings, significantly reduce carbon emissions, and effectively address the challenges faced by rural energy development. This study explores a typical framework for rural MECS that integrates photovoltaic, wind turbine, and biomass biogas combined cooling, heating, and power technology while considering the partial load ratio of equipment components and coupling characteristics between different energy sources. Based on various scenarios of valley electricity utilization, multi-objective optimization models are established to determine the capacity of MECS with economy, environment, and primary energy saving rate as objective functions. The non-dominated sorting genetic algorithm (NSGA-II) along with Technique for Order Preference by Similarity to Ideal Solution decision-making method is adopted to obtain optimal solutions from the Pareto solution set. The case study conducted in a rural area of central China has demonstrated the effective enhancement of coupling capacity in MECS through battery storage. By actively storing energy during off-peak electricity periods, battery storage strengthens the complementary capabilities of photovoltaic systems, wind turbines, and itself. This approach allows for a reduction in planned capacity for photovoltaic and wind power systems within MECS while increasing the planned capacity for internal combustion engines, resulting in respective decreases in system investment costs by 16.19% and 13.18%. Furthermore, incorporating more biogas-fired cogeneration during off-peak electricity periods improves the system's performance economically, environmentally, and with regards to primary energy saving rate.

1. Introduction

In line with China's goal of carbon peaking and carbon neutrality, a new energy strategy has been proposed and implemented, making renewable energy the cornerstone of China's energy system [1]. The promotion of sustainable development in renewable energy and the implementation of guiding policies for rural revitalization in China are leading to significant transformations in the rural energy system [2]. In the past, rural households mainly relied on public utility power grids for electricity and used various sources like electricity, straw, firewood, and coal for heating. Some households installed solar collectors to partially meet their heating needs. However, with the development of clean energy and the implementation of the rural revitalization strategy in China, renewable energy systems based primarily on solar and wind power have been widely promoted. The utilization efficiency of biomass energy such as straw and firewood can be significantly improved in rural areas due to advanced agriculture practices and abundant availability of biomass resources. This can be achieved through technologies like biogas cogeneration. Currently, coupling and utilizing various forms of renewable energy to establish a multi-energy complementary system (MECS) is crucial for addressing challenges like low rural energy utilization efficiency and environmental pollution while meeting the living and production needs of rural areas [3]. Nevertheless, achieving complementarity among diverse renewable energy sources remains challenging due to their unique characteristics. Additional research is required regarding the coupling utilization plan of rural renewable energy in comparison with urban areas.

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Nomenclature

AC/H	Absorption chiller/heater
AM	Atmospheric acceptance of sunlight
AM_0	Acceptance under standard operating conditions
BS	Battery storage
C_{fc}	Equipment maintenance costs
$C_{fc,i}$	maintenance cost of equipment i
C_{gb}	Electricity purchase costs from the public grid
C_{inv}	Investment and construction cost of MECS
$C_{inv,i}$	Installation and construction cost per unit capacity of equipment i
C_{opt}	Operation cost of MECSInitial
$C_{price,t}$	Real-time electricity price
C_{toal}	Life cycle cost of the MECS
CCHP	Combined cooling, heating, and power
CE	Total carbon emissions over a period
$Cell_{BS}$	Capacity of the battery storage device
$Cell_{TS}$	Capacity of the thermal storage device
ch	Charge state of the energy storage equipment
CHPB	Cogeneration system of heat, power and biogas
$COP_{AC/H}$	Rated coefficient of performance of the absorption chiller/heater
COP_{GSHP}	Rated coefficient of performance of ground source heat pump
d	Discharge state of the energy storage equipment
E_{grid}	Electricity purchased from the grid by MECS system
E_{gridcon}	Electricity purchased from the grid by traditional SP system
$E_{k,t}$	Remaining energy in the storage device at time t
$E_{k,t+1}$	Remaining energy in the storage device at time $t+1$
EB	Electric boiler
em_{bio}	Carbon emission coefficient of biomass biogas
em_g	Carbon emission factor of the regional grid
ES	Electrical storage
F_{ng}	Gas consumed by the MECS system
$F_{\text{ng,co}}$	Gas consumed by traditional SP system
F_R	Heat removal coefficient
FEL	Follow the electric load
G	Solar radiation intensity
G_0	Solar radiation intensity under standard operating conditions
GSHP	Ground source heat pump
HR	Heat recovery unit
i	Different equipment in the system
I	Set of all the equipment
ICE	Internal combustion engines
IES	Integrated Energy System
LHV	Lower heating value of biogas
M_{fuel}	Total mass of straw consumed
MO-NSGA-II	multi-objective non-dominated sorting genetic algorithm
MECS	Multi-energy complementary systems
NSGA-II	Non-dominated sorting genetic algorithm
$P_{AC/H,in}$	Rated capacity of the absorption chiller
p_{b0}	Price per unit mass of straw
$P_{bio,in}$	Rated capacity of the biogas engine
$P_{GSHP,in}$	Capacity of the GSHP
$P_{in,i}$	Design capacity of equipment i
p_{t}	Unit transportation price
$P_{k,t}^{chr}$	Charging power of the energy storage device at time t
$P_{k,t}^{dis}$	Discharging power of the energy storage device at time t
$P_{GSHP,e}^{in}$	Input electricity to the GSHP
$P_{bio,e}^{out}$	Electricity output of the biogas engine
$P_{bio,h}^{out}$	Heat load provided by the biogas cogeneration system
P_{grid}^{out}	Purchased grid electricity
$P_{i,t}^{out}$	Output of equipment i at time t
$P_{PV,e}^{out}$	Photovoltaic output
PESR	Primary energy saving rate
PLR	Partial load ratio
PLR_{AC}	effective load coefficient of AC/H
PLR_{bio}	Partial load ratio of biogas engine
PLR_{GSHP}	Partial load ratio of ground source heat pump
PV	Photovoltaic
$Q_{AC,h}^{in}$	Heat input to the chille
$Q_{bio,h}^{in}$	Heat generated by the consumption of biogas
$Q_{AC,c/h}^{out}$	Cooling/Heating output of the absorption chiller
$Q_{bio,h}^{out}$	Heat output of the biogas engine
$Q_{GSHP,h/c}^{out}$	Output of the GSHP
$Q_{SC,h}^{out}$	Output of the solar collector
r	Depreciation rate
R	Straw collection radius
S_i^+	Distance from the target to the positive ideal solution ν_j^+
S_i^-	Distance to the negative ideal solution ν_j^-
S_{PV}	Installed photovoltaic panel area
S_{SC}	Installed solar thermal collector area
SC	Solar thermal collector
$SOC_{k,t}$	State of charge of the k-type energy storage device at time t
$SOC_{k,t+1}$	State of charge of the k-type energy storage device at time t + 1
SOC_k^L	Lower bound of the energy storage equipment state
SOC_k^U	Upper bound of the energy storage equipment state
SP	Separate production
T	Outdoor temperature
T_{in}	Inlet water temperature of the heat collector
T_0	Temperature under standard operating conditions
TOPSIS	Technique for Order Preference by Similarity to Ideal Solution
TS	Thermal storage
V	Actual wind velocity
$V_{biogas,t}$	Volume of biogas
$V_{\text{cut-in}}$	Cut-in wind velocity
$V_{\text{cut-out}}$	Cut-out wind velocity
$V_{in,t}$	Inlet power of biogas
$V_{out,t}$	Outlet power of biogas
V_r	Rated wind velocity
WT	Wind turbine
X	Decision variables
x_i	Equipment's life cycle
α_0	Heat loss coefficient
β	Distance tortuosity factor
LCA	Life cycle assessment
β_{i}	Carbon emission coefficient of equipment i
ρ	Straw resource density
$\gamma_{k,d}^{max}$	Upper limits of single discharge of the energy storage equipment
$\gamma_{k,ch}^{max}$	Upper limits of single charge of the energy storage equipment
$\mu_{k,t}^{che}$	Variable in charging discharging states
$\mu_{k,t}^{dis}$	Variable in discharging states
ϑ_j^+	Positive ideal solution for the objective j
ϑ_j^-	Negative ideal solution for the objective j
η_{bio}^E	Power generation efficiency of the biogas engine
η_{bio}^{loss}	Heat loss rate of the biogas engine
η_{bio}^Q	Heat generation efficiency of the biogas engine

η_{e}	Transmission efficiency of the grid	$\eta_{k,dis}$	Discharging efficiencies
η_{grid}	Power generation efficiency of the power plant	η_{PV}	Energy conversion efficiency
η_{hr}	Heat recovery efficiency	$\eta_{SC,0}$	Instantaneous heat collection and interception efficiency of the solar collector
η_{hx}	Efficiency of the heat exchanger	η_{SC}	Collector efficiency
$\eta_{k,chr}$	Charging efficiencies		

To explore a clean and efficient energy system for rural areas, researchers have conducted studies on energy consumption characteristics, renewable energy utilization patterns and planning and operation strategies of MECS in rural areas. Wu et al. [4] quantitatively described the energy consumption patterns of rural households in China and investigated consumption quantity, fuel structure and demand structure across different regions. Zou and Li et al. [5,6] comprehensive ly surveyed energy consumption data in Chinese rural households and analyzed the main factors influencing their energy usage. Han et al. [7] studied the spatial–temporal characteristics of the energy ladder in Chinese agriculture and rural life using a spatial–temporal model. Chen et al. [8] investigated the transition process from traditional biomass to cleaner and more efficient biomass energy, along with factors affecting energy choice behavior. Tan et al. [9] investigated the feasibility of implementing MECS in rural areas across central, eastern, and western China, focusing on economic viability, energy conservation, and social benefits. These studies have explored issues related to rural household energy consumption and the utilization of renewable energy sources. Currently, there is an imbalance between energy supply and consumption in rural China due to the traditional methods that suffer from high environmental pollution and low energy efficiency. However, establishment MECS can effectively harness renewable energy sources such as solar energy, biomass energy, and geothermal energy while improving utilization efficiency and providing economic benefits for rural residents.

Researchers have studied various aspects in the planning of MECS, such as exploring resource coupling utilization, optimizing objectives using different indicators, and developing scheduling strategies. Regarding resource coupling, Xu et al. [10] introduced a geothermal-solar-wind MECS framework for community multi-energy supply. They also developed a scheduling matrix to study microgrid complementarity in both grid-connected and off-grid modes. Tang et al. [11] investigated optimal capacity and operation modes for complementary photovoltaic-wind power generation systems. Sun et al. [12] examined how wind-solar hybrid energy systems can be effectively integrated into distribution networks through their complementarity. Luz et al. [13] optimized the integration of batteries by considering the complementarity between source-area, costs, and demand-side management.

In terms of scheduling strategies, Zhong et al. [14] proposed an MECS management strategy that combines day-ahead and intra-day scheduling to consider costs and carbon emissions. This strategy effectively reduces operating costs and carbon emissions. The allocation of system capacity and energy scheduling strategies were coordinated and optimized using a multi-objective mixed integer linear programming method by Chen et al. [15]. Zou et al. [16] proposed a dynamic optimization scheduling factor coupling matrix based on the synergistic interaction of electricity, gas, and heating energy flows, as well as the thermodynamic effect of digestion for multi-carrier power generation scheduling. This model simulates the production, conversion, storage, and consumption of various energy carriers and has been tested on an independent microgrid within a 24-hour scheduling range.

In terms of optimization indicators, Liu et al. [17] considered the impact of uncertainty and established a capacity optimization model for an integrated energy system that utilizes solar energy and gas in combination. Using non-dominated sorting genetic algorithm (NSGA-II), they optimized the model by considering economic, environmental, and thermal comfort as the objective functions. Wang et al. [18] optimized system capacity optimization focusing on economic, environmental, and energy benefits. Marek et al. [19] optimized the configuration of integrated energy systems for various hybrid energy scenarios, take into account economic and environmental factors, and analyzed the relationship between optimization objectives. Xu et al. [20] proposed a method that combines orthogonal design and genetic algorithm to optimize the capacity of equipment within an integrated energy system, resulting in significantly reducing investment costs after optimization. Wang et al. [21] considered the uncertainty of loads and renewable energy sources, represented the uncertainty of solar irradiance and building loads using parametric methods of probability distribution, and optimized a hybrid combined cooling, heating, and power system using NSGA-II to achieve optimal energy, economic, and environmental benefits. Wu et al. [22] conducted a study on a multi-functional park and developed a multi-objective optimization model that integrates operating costs, energy efficiency, and penalties for polluting emissions. The multi-objective non-dominated sorting genetic algorithm (MO-NSGA-II) algorithm was proposed to optimize the model. Yang et al. [23] established an optimization model for capacity planning in an integrated energy system that utilizes wind, solar, and geothermal energy, with the objective functions of reliability, economy, and environmental protection. They also investigated the impact of scheduling strategies and grid penetration rates on the integrated energy system. The system's reliability can be greatly improved by considering it in the optimization model, with only a small sacrifice in environmental performance and economy. Lin et al. [24] proposed a quantitative analysis of the complementarity index for integrated energy systems that satisfying regional cooling, heating, and power loads. They considered operational, investment, and maintenance costs of renewable energy sources and studied the scheduling patterns and configurations for long-term planning of regional integrated energy systems. The results showed that the quantitative relationship between complementarity and economy, in terms of both initial investment and long-term scale, helps propose optimal system configuration strategies, effectively alleviating peak loads on the power grid.

Crop production in rural areas is abundant, and rural areas have tremendous biomass potential compared to urban areas. Utilizing crop by-products to generate biogas through anaerobic digestion is an effective means of harnessing biomass energy. As a renewable and combustible gas, biogas is considered one of the most promising substitutes for natural gas due to its high energy density, widespread availability of raw materials, and abundant reserves. Extensive research has been conducted on it in recent years. Zhang et al. [25] constructed an environmental benefit index model for the cogeneration system of heat, power and biogas (CHPB), considering dynamic climate change, energy consumption characteristics, and environmental parameters, and analyzed the environmental effects of the system using equivalent emission factors. Moshood et al. [26] investigated the motivation for the startup of a microgrid dominated by biogas cogeneration. Towhid et al. [27] examined the thermodynamic and economic advantages of biogas cogeneration. Zhang et al. [28] quantified the monetary value of the environmental effects generated by biogas projects and evaluated the economic impacts of these projects using the life cycle assessment (LCA) method. Jun et al. [29] investigated the utilization of biogas and concluded that biogas cogeneration is more economically efficient than combined cycles when the electricity consumption resulting from annual heat demand accounts for 30 % of the total annual electricity consumption.

With the increasing energy demand in rural areas, a single form of energy has gradually become insufficient to meet user needs. Therefore, establishing a multi-energy complementary energy system has become an important approach to addressing rural energy issues and achieving rural revitalization. Through a multi-energy complementary energy system, the complementarity among different energy sources can be harnessed to enhance the stability and reliability of energy supply, thereby contributing to the sustainable and dependable operation of the system. Numerous studies have demonstrated that multi-energy complementary systems, which incorporate photovoltaics and biomass, outperform independent biomass systems in terms of both economy and environment. Currently, researchers are including biogas in the multi-energy complementary system for rural areas for further investigation.

Zhi et al. [30] investigated the electricity usage of different types of households in a rural area, studying various management strategies for photovoltaic (PV) battery systems. With the objectives of achieving economic efficiency and ensuring smooth grid interaction, he optimized the scenarios and explored the possibility of applying power plant optimization strategies and evaluation methods to rural power system design. Li et al. [31] used a village in western China as an example to demonstrate the technical and economic feasibility of an off-grid hybrid renewable energy system for remote rural electrification. Through system simulation, he identified the most cost-effective configuration and compared the off-grid hybrid power system with grid expansion. The results indicated that a hybrid system, which combines solar, wind, and biomass energy, is a reliable and cost-effective choice for achieving sustainable rural electrification in remote areas. Li et al. [32], focused on a village in Xuzhou, eastern China, and established an optimized hybrid energy system that combines photovoltaic, biogas, diesel, and batteries to meet the electrical load of the village. The optimized results showed that the PV-biogas hybrid energy system had the best economic effect, while the PV-battery hybrid energy system was the most environmentally friendly, emitting no pollutants. On the other hand, the diesel generator system had the highest emissions of pollutants. Chen et al. [33] focused on heat demand and proposed a centralized hybrid heating system that combines solar and biogas to meet the heating load in rural areas. They developed a simulation model to calculate and analyze the energy and thermodynamic performance of the hybrid heating system, as well as investigate the impact of capacity parameters of its components on its thermodynamic performance. The results showed that the hybrid heating system had superior energy-saving and economic performance compared to traditional single solar heating systems. Qin et al. [34] considered thermal comfort and fluctuations in the price of natural gas. They used penalty factors to describe thermal comfort and a dynamic Bayesian network model to predict natural gas fluctuations. They established a two-level optimization model for an integrated biogas energy system, which improved energy agent profits and equipment flexibility towards the main output unit, effectively reducing dependence on the power grid and gas grid. Wang et al. [35] proposed a planning method for integrated biogas-solar energy systems in rural areas and an optimization strategy based on rural scenarios. They constructed a planning optimization model with the objectives of optimal energy efficiency, economic optimality, and lowest carbon emissions, and optimized it using the winter period as an example. Fan et al. [36] established an integrated energy model, including biogas cogeneration, based on the characteristics of rural natural resources. They optimized the winter scheduling with the objectives of minimizing energy operation costs and carbon emissions while maximizing energy satisfaction.

In summary, current research primarily focuses on integrated energy systems planning in industrial parks and commercial districts, while research on such systems in rural areas mainly discusses development, perspectives and planning frameworks. There are fewer studies on comprehensive energy systems that fully utilize biomass and solar energy, requiring futher investigation into co-supply models like biogas cogeneration. Additionally, most existing researches on the design and planning of integrated energy systems in rural areas focus solely on either electrical or thermal energy flow as the core, with limited attention given to both simultaneously. There is a lack of studies addressing simultaneous cooling, heating, hot water, and power requirement. Compared to industrial parks, existing researches on the design and planning of integrated energy systems in rural areas does not fully utilize time-of-use electricity price differences nor involve sufficient participation of energy storage equipment. The impact of component load ratios is often neglected when constructing system models incorporating biogas engines. Therefore, this study combines the characteristics of rural resource endowments to conduct researches on MECS planning based on biomass and solar energy resources with tha aim of accelerating its application in rural areas. The specific contributions of this research are as follows:

1) A typical rural Multi-Energy Complementary System (MECS) framework has been developed, integrating photovoltaic (PV), wind turbine (WT), and biomass biogas combined cooling, heating, and power (CCHP) technology. Furthermore, a rural MECS planning model has been established with the objectives of minimizing life-cycle costs, reducing carbon emissions, and maximizing primary energy savings.
2) A scheduling strategy has been proposed for rural MECS, which describes the coordinated operation between the electrical system and the cooling/heating devices. This strategy considers energy storage methods (electrical, thermal, and biogas) as well as the partial load ratio (PLR) of each equipment component instead of using constant efficiency, providing a more realistic simulation.
3) The study examines the performance of the MECS under various operation modes, taking into account the cost of biogas feedstock, collection and transportation.

2. System description

2.1. Structure of rural multi-energy complementary system

The aim of this research is to develop a rural multi-energy system that integrates biomass, solar, geothermal energy, and the public grid. This system will meet the daily electricity, cooling/heating, and hot water needs of rural households in different regions by fully utilizing renewable energy resources in areas with fewer high-energy-consuming buildings. The fundamental framework of the rural multi-energy complementary system is illustrated in Fig. 1. Biogas cogeneration technology, photovoltaic arrays, wind turbine and the public grid are used to meet electricity demand in rural areas. Solar thermal technology and ground-source heat pumps are employed for cooling, heating and hot water demands. Waste heat generated by biogas cogeneration is utilizedd by heat recovery devices and absorption chillers to provide collaborative cooling and heating solutions. Thermal and electrical energy storage devices mitigate fluctuations in MECS and improve energy utilization efficiency. Additionally, the effective PLR impact is considered due to controllable renewable energy equipment not always operating at rated power. The following section will discuss the specific configurations within the model.

2.2. Mathematical models of devices

2.2.1. Photovoltaic array

PV systems receive solar radiation and convert it into electrical energy. The amount of electricity generated depends on the radiation intensity, PV system area and energy conversion efficiency. It is calculated using Eq. (1) [37].

$$P^{out}_{PV,e} = GS_{PV}\eta_{PV} \tag{1}$$

Where, *Pout PV,e* represents the photovoltaic output, kW; G represents the solar radiation intensity, kW; S_{PV} represents the installed photovoltaic panel area, m^{-2}; η_{PV} represents the energy conversion

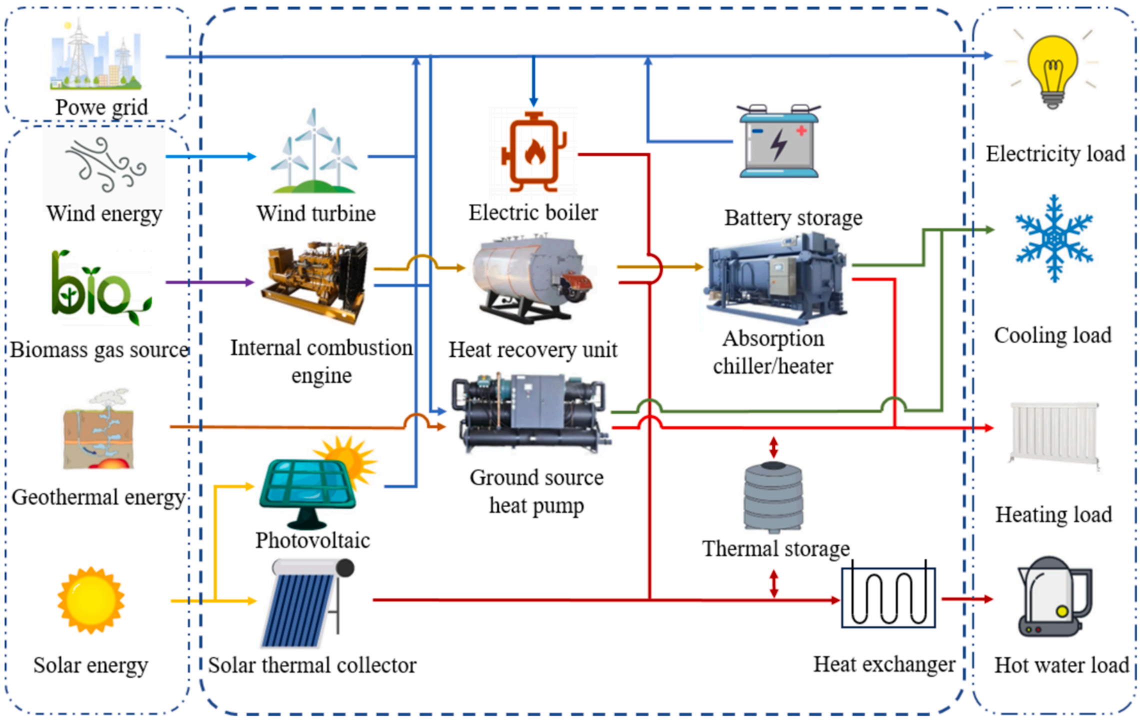


Fig. 1. Schematic diagram of the MECS.

efficiency, which is related to the solar radiation intensity, working environment temperature, and atmospheric air quality, and is calculated according to Eq. (2) [38].

$$\eta_{PV} = k_1\left[\left(\frac{G}{G_0}\right)^{k_2} - k_3\left(\frac{G}{G_0}\right)\right][1 - k_4\left(\frac{T}{T_0}\right) + k_5(\frac{AM}{AM_0})] \tag{2}$$

Where, G_0 is the solar radiation intensity under standard operating conditions, kW/m^{-2}, with a value of 1; T is the outdoor temperature; T_0 is the temperature under standard operating conditions, with a value of 25℃; AM represents the atmospheric acceptance of sunlight, kg; AM_0 is the acceptance under standard operating conditions, with a value of 1.5.

2.2.2. Solar thermal collector

The solar thermal collector (SC) receives solar radiation and converts it into heat energy. Similar to PV components, the efficiency of solar heat collection is mainly restricted by meteorological parameters such as ambient temperature and solar radiation intensity, as well as the parameters of the collector itself. It is calculated according to Eq. (3) and Eq. (4) [37].

$$Q^{out}_{SC,h} = GS_{SC}\eta_{SC} \tag{3}$$

$$\eta_{SC} = F_R[\eta_{SC,0} - \frac{\alpha_0(T_{in} - T)}{G}] \tag{4}$$

Where, $Qout\ SC,h$ represents the output of the solar collector, kW; S_{PV} represents the installed solar collector area, m^{-2}; F_R is the heat removal coefficient; η_{SC} is the collector efficiency; $\eta_{SC,0}$ is the instantaneous heat collection and interception efficiency of the solar collector; α_0 is the heat loss coefficient, kW/(m^2•℃); T is the outdoor ambient temperature, ℃; T_{in} is the inlet water temperature of the heat collector, ℃.

2.2.3. Wind turbine

The wind turbine (WT) output model is generally described by Eq. (5) [39]:

$$P^{out}_{WT,e} = \begin{cases} 0, V < V_{cut-in}, V > V_{cut-out} \\ \dfrac{V^3 - V_{cut-in}{}^3}{V_r{}^3 - V_{cut-in}{}^3} \cdot P_{WT,in}, V_{cut-in} \le V < V_r \\ P_{WT,in}, V_r \le V < V_{cut-out} \end{cases} \tag{5}$$

Where, $Pout\ WT,e$ represents the wind turbine output, kW; $P_{WT,in}$ represents the installed capacity of the wind turbine, kW; $V_{\text{cut-in}}$, $V_{\text{cut-out}}$, V and V_r represent the cut-in wind velocity, cut-out wind velocity, actual wind velocity and rated wind velocity, respectively, m/s.

2.2.4. Biogas thermoelectric cogeneration

2.2.4.1. Internal combustion engine. The internal combustion engine burns biogas as input to generate electricity and heat, and the model is as follows Eq. (6) and Eq. (7) [40]:

$$P^{out}_{bio,e} = Q^{in}_{bio,h}\eta^E_{bio} = V_{in,t}LHV\eta^E_{bio} \tag{6}$$

$$P^{out}_{bio,h} = Q^{in}_{bio,h}\eta^Q_{bio} \tag{7}$$

Where, $Pout\ bio,e$ represents the electricity output of the biogas engine, kW; $Qin\ bio,h$ represents the heat generated by the consumption of biogas, $Pout\ bio,h$ represents the heat output of the biogas engine, $V_{in,t}$ represents the volume of input biogas, m^3; LHV represents the lower heating value of biogas, kJ/m^3; $\eta E\ bio$, $\eta loss\ bio$and $\eta Q\ bio$ represent the power generation efficiency, heat loss rate, and heat generation efficiency of the biogas engine, respectively. They are treated as functions of the effective partial load ratio (PLR_{bio}) during operation and calculated according to Eq. (8) and Eq. (9) [41].

$$\eta^E_{bio} = 0.1841 + 0.1736PLR_{bio} + 0.0102PLR^2_{bio} + 0.0122PLR^3_{bio} \tag{8}$$

$$\eta^Q_{bio} = 1 - \eta^E_{bio} - \eta^{loss}_{bio} \tag{9}$$

The effective part-load ratio of the biogas engine is defined as Eq. (10):

$$PLR_{bio} = \frac{P^{out}_{bio,e}}{P_{bio,in}} \tag{10}$$

Where, $P_{bio,in}$ is the rated capacity of the biogas engine, kW.

2.2.4.2. Heat recovery. The heat recovery unit is used to recover the waste heat from the biogas engine, and the calculation is as Eq. (11) [20].

$$Q^{out}_{hr,h} = P^{out}_{bio,h}\eta_{hr} \tag{11}$$

Where, η_{hr} represents the heat recovery efficiency.

The heat load is provided by the heat exchanger, and the final heat load provided by the biogas cogeneration system [20]:

$$Q^{out}_{bio,h} = Q^{out}_{hr,h}\eta_{hx} \tag{12}$$

Where, *Pout bio,h* represents the heat load provided by the biogas cogeneration system, kW; η_{hx} represents the efficiency of the heat exchanger.

2.2.4.3. Biogas generator. Biogas generators produce biogas through anaerobic reactions. The biogas generator mentioned in this research has a volume of 10 m^3, and each unit produces biogas at a rate of 0.2 m^3/h. This research focuses on the number of biogas generators, and specific details regarding biogas production are not within the scope of this study. Furthermore, the biogas generator also serves as a storage device for biogas, and the volume of biogas stored in the generator must not fall below the set minimum value.

$$V_{biogas,t} = V_{biogas,t-1} + V_{in,t} - V_{out,t} \tag{13}$$

Where, $V_{biogas,t}$ represents the volume of biogas, $V_{in,t}$ represents the inlet power of biogas, $V_{out,t}$ represents the outlet power of biogas.

2.2.5. Absorption chiller/heater

The absorption chiller/heater (AC/H) system excels in utilizing low-grade heat energy and waste heat. In this system, the waste heat generated during the operation of the biogas engine can be utilized to provide both cooling and heating loads. The cooling and heating coefficients are related to the effective part-load ratio [42].

$$Q^{out}_{AC,c/h} = Q^{in}_{AC,h}COP_{AC/H} \tag{14}$$

$$\text{For cooling}: COP_{AC} = 0.425 + 1.683PLR_{AC} - 2.419PLR^2_{AC} + 1.108PLR^3_{AC} \tag{15}$$

$$\text{For heating}: COP_{AH} = COP_{AH,ra} \tag{16}$$

$$PLR_{AC/H} = \frac{Q^{out}_{AC,c/h}}{P_{AC/H,in}} \tag{17}$$

Where *Qout AC,c/h* represents the cooling/heating output of the absorption chiller, kW; *Qin AC,h* represents the heat input to the chiller, kW; $COP_{AC/H}$ is the actual coefficient of performance of the absorption chiller/heater, $COP_{AH,ra}$ is the rated coefficient of performance of the absorption chiller/heater, PLR_{AC} is the effective load coefficient of AC/H, $P_{AC/H,in}$ is the rated capacity of the absorption chiller, kW.

2.2.6. Ground source heat pump

The ground source heat pump (GSHP) system can provide cooling/heating loads, which are calculated using the following Eq. (18) –Eq. (20) [43].

$$Q^{out}_{GSHP,h/c} = P^{in}_{GSHP,e}COP_{GSHP} \tag{18}$$

$$PLR_{GSHP} = \frac{Q^{out}_{GSHP,h/c}}{P_{GSHP,in}} \tag{19}$$

$$\frac{COP_{GSHP}}{COP_{GSHP,ra}} = (0.833 + 0.197PLR_{GSHP} - 0.038PLR^2_{GSHP} + 0.007PLR^3_{GSHP}) \tag{20}$$

Where, *Qout GSHP,h/c* represents the output of the GSHP, kW; *Pin GSHP,e* is the input electricity to the GSHP, kW; $P_{GSHP,in}$ is the capacity of the GSHP, kW; PLR_{GSHP} is the PLR of the effective component, defined as the ratio of actual output power to rated power; COP_{GSHP} is the rated coefficient of performance of the GSHP, calculated based on the PLR_{GSHP} and the rated $COP_{GSHP,ra}$.

2.2.7. Energy storage

Energy storage devices include electrical storage (ES) and thermal storage (TS), whose models are described as Eq. (21)- Eq. (23) [24].

$$SOC_{k,t} = \frac{E_{k,t}}{Cell_k} \tag{21}$$

$$E_{k,t+1} = E_{k,t} + \mu^{chr}_{k,t}\eta_{k,chr}P^{chr}_{k,t} - \mu^{dis}_{k,t}\eta_{k,chr}P^{dis}_{k,t} \tag{22}$$

$$SOC_{k,t+1} = SOC_{k,t} + \frac{\mu^{chr}_{k,t}\eta_{k,chr}P^{chr}_{k,t} - \mu^{dis}_{k,t}\eta_{k,dis}P^{dis}_{k,t}}{Cell_k} \tag{23}$$

Where, k represents the type of energy storage device; $E_{k,t}$ and $E_{k,t+1}$ represent the remaining energy in the storage device at time t and time $t+1$, respectively, kWh; $SOC_{k,t}$ and $SOC_{k,t+1}$ are defined as the state of charge of the k-type energy storage device at time t and time t + 1, respectively; *Pchr k,t* and *Pdis k,t* represent the charging and discharging power of the energy storage device at time t respectively, kW; *μchr k,t* and *μdis k,t* are 0–1 variables, indicating the charging and discharging states, respectively; $\eta_{k,chr}$ and $\eta_{k,dis}$ are the charging and discharging efficiencies, respectively ; $Cell_k$ is the capacity of the energy storage device, kWh.

2.3. Operation strategy

2.3.1. Electrical operation strategy

As depicted in Fig. 2, the electrical load required by users is supplied jointly by uncontrollable renewable energy generation devices and controlled renewable energy devices. Solar energy and wind energy are renewable energy sources that cannot be controlled, and their power output is greatly influenced by meteorological conditions. The prioritization of PV and WT is to provide electricity. When PV and WT meet the load requirements and have surplus power, ES will store the excess electricity and sell it to the grid. If PV and WT still cannot meet the required load, consideration will be given to operating the ICE. As the system operates in practice, the operation of ICE is determined by its PLR, and its efficiency is related to its PLR. If the PLR of the ICE is too low during operation, it will result in inefficient operation of the ICE. In this study, it is set that the ICE will be turned on when its PLR exceeds 0.4 and will not be activated when it falls below 0.4 [44]. Based on this, ES will serve as a supplement to the ICE by providing power when the ICE is inactive or unable to meet the user's power demand, thereby minimizing the amount of electricity purchased from the grid during non-valley period. In addition, the operation strategy of the ICE follows the follow the electric load (FEL) mode, which is based on the core principle of not generating excessive power. Additionally, apart from meeting the basic load, the system also needs to provide power for the requirements of GSHP. In cases where none of these devices can meet the required electrical load, any shortage of power will be supplied by the national public grid. Finally, during valley period when electricity prices are low and if the electrical storage equipment is not at full capacity, it will be charged to enhance its regulation capability.

2.3.2. Heating/Cooling/Hot water operation strategy

The heating/cooling demands and hot water needs of users are

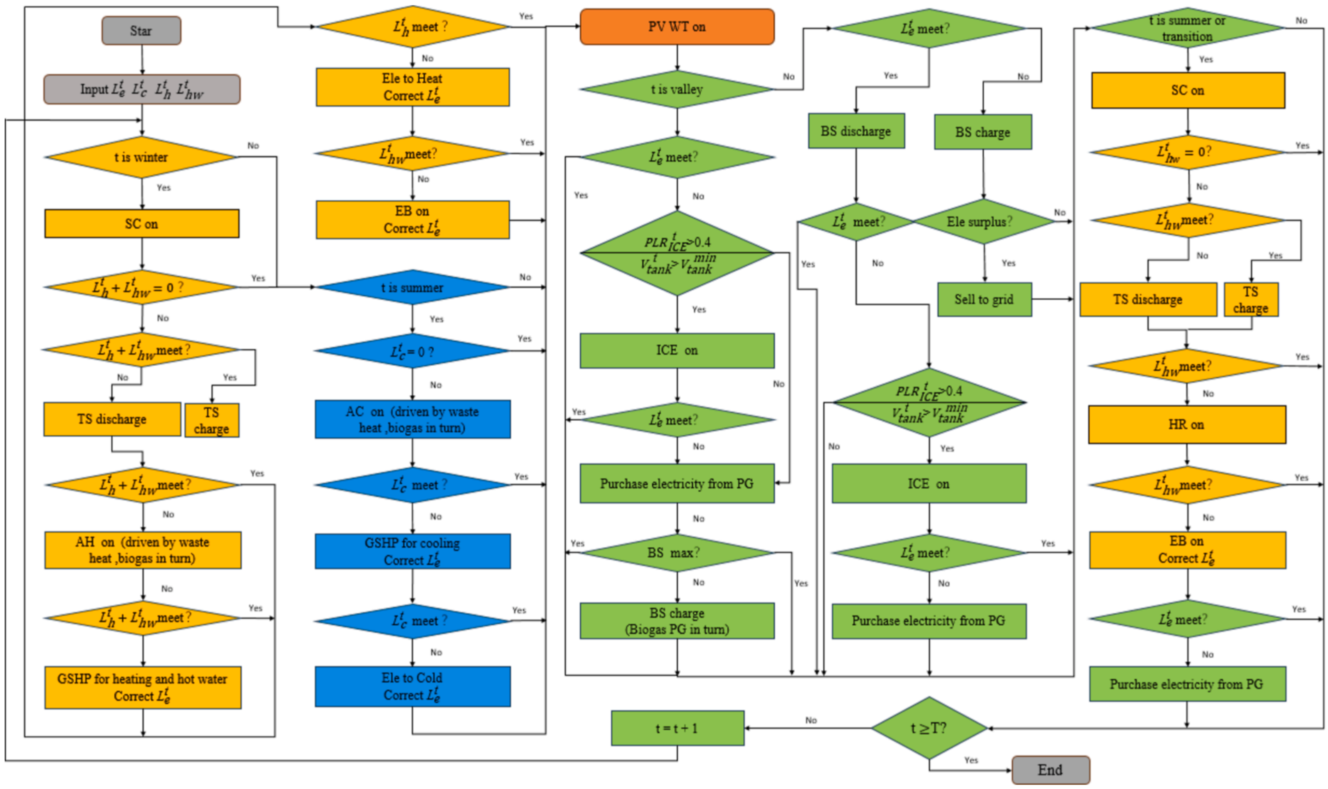


Fig. 2. Scheduling strategy for the MECS.

primarily provided by SC, GSHP, and AC/H. In winter, the heat collected by SC is stored in the hot water storage tank. The heating and hot water demands of users are first supplied by SC and TS. If the heating load and hot water load cannot be fully met, AH will be activated, and WHR will absorb the heat from the exhaust gas generated by ICE to provide heat to AH. If AH still cannot satisfy the heating and hot water demands of users, GSHP will be turned on to provide the remaining required heat with priority given to meeting user's hot water needs. Unsatisfied hot water demands will be met by electric boiler (EB). Additionally, in summer and transition seasons when there is no heating load but only a demand for hot water mainly provided by SC and heat recovery unit (HR), during these times when the demand for hot water is relatively small, the system can basically meet user's requirements for it. For cooling loads, AC is used initially to provide cooling through absorbing of recovered heat from HR. If AC fails to meet user's cooling demand then GSHP provides insufficient cooling capacity. The input electrical load will be counted as part of electrical consumption with would also be provided by this system. Unsatisfied cooling demands would then have electric cooling as an alternative solution.

3. Methodology

Based on the framework model and scheduling strategy of the rural multi-energy complementary system introduced in the previous section, this section aims to establish a multi-objective optimization model for rural multi-energy complementary. Considering the relatively low income in rural areas, it is necessary to take into account the economic benefits of the MECS to avoid increasing the residents' financial burden. Traditional methods of energy utilization in rural areas generate significant carbon emissions, while the MECS system offers substantial improvements in this aspect. Therefore, it is essential to consider the carbon emissions of the system as environmental benefits. Additionally, traditional rural energy sources tend to have low energy efficiency, which can be significantly improved by implementing an MECS. The primary energy saving rate will be used as a measure to represent the degree of improvement offered by an MECS. This study will construct a planning model for rural MECS that comprehensively considers economic benefits, environmental benefits, and primary energy saving rate, as shown in Fig. 3.

3.1. Objective function

3.1.1. Economic objectives

In this study, the minimum life cycle cost is used as an economic indicator. The life cycle cost of the MECS consists of two parts [28]: initial investment and construction cost C_{inv} and system operation cost C_{opt}.

$$C_{total} = C_{inv} + C_{opt} \tag{24}$$

Specifically, C_{inv} includes construction costs and equipment purchase costs, calculated as Eq. (25):

$$C_{inv} = \sum_{i \in I} \frac{r(1+r)^{x_i}}{(1+r)^{x_i} - 1} c_{inv,i} P_{in,i} \tag{25}$$

Where, I represents the set of all equipment; i represent different equipment in the system; r represents the depreciation rate, which is set at 8 % in this study; x_i represents the equipment's life cycle; $C_{inv,i}$ is the installation and construction cost per unit capacity of equipment i; $P_{in,i}$ represents the design capacity of equipment i; Additionally, the capacities of PV and SC are controlled by S_{PV} and S_{SC}.

The operation cost C_{opt} is composed of equipment maintenance costs C_{fc} and electricity purchase costs from the public grid C_{gb}, calculated as Eq. (26):

$$C_{opt} = C_{fc} + C_{gb} \tag{26}$$

Equipment maintenance cost C_{fc}:

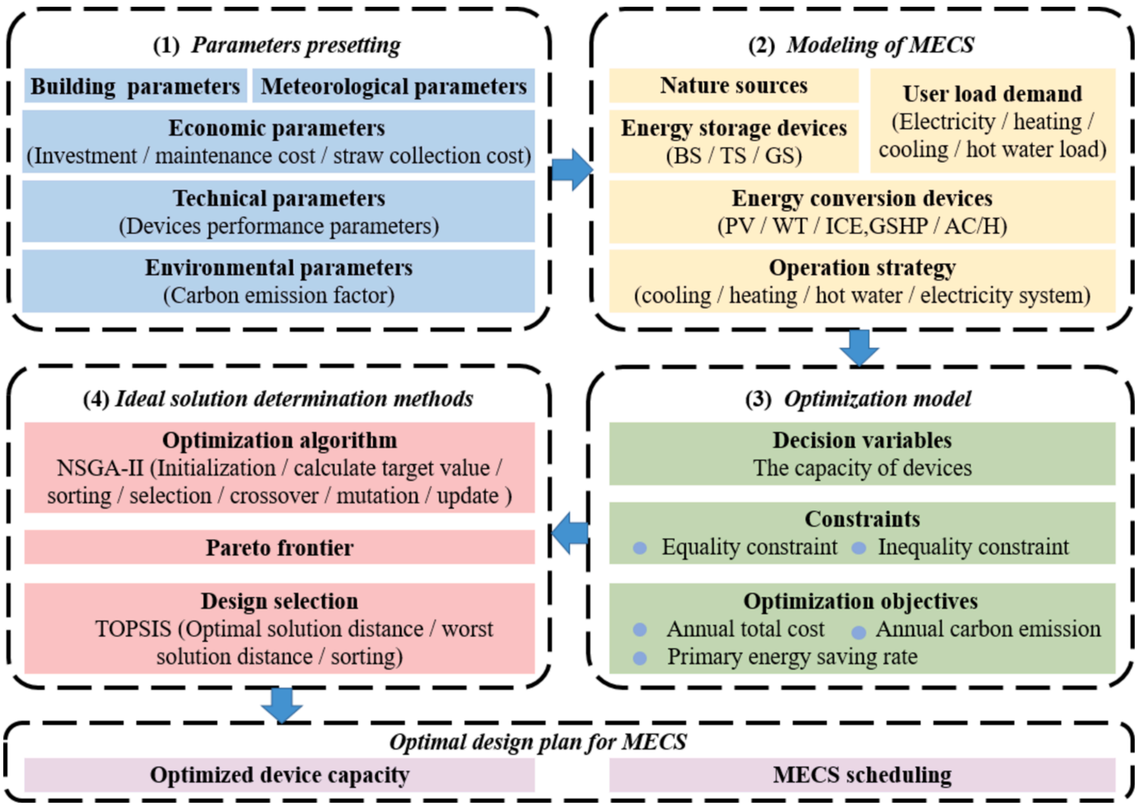


Fig. 3. Optimization process of the methodology.

$$C_{fc} = \sum_{i \in I} \sum_{i=1}^{8760} P_{i,t}^{out} c_{fc,i} \tag{27}$$

Where *Pout i,t* is the output of equipment *i* at time t, kW; $C_{fc,I}$ is the maintenance cost of equipment *i*, $/kW.

Electricity purchase cost from the grid C_{gb}:

$$C_{gb} = \sum_{i=1}^{8760} P_{grid}^{out} c_{price,t} \tag{28}$$

Where *Pout grid* is the purchased grid electricity, kWh; $C_{price,t}$ is the real-time electricity price, $/kWh.

Due to the relatively simple population distribution, transportation conditions, and energy demand in rural areas, typically only one biomass power plant is established in a specific region. The main raw material for biogas cogeneration technology is straw from nearby farmland. This study adopts a centralized transportation and storage model [45,46] for the collection, transportation,and storage of straw. This model has a simple operation principle with fewer intermediate links and relatively uncomplicated straw processing methods, making it suitable for small-scale biomass power plants in a given region.

To simplify the model, the following assumptions are made regarding storage and transportation costs:

1) The straw fuel in the collection area is distributed uniformly with the same density.
2) The straw collection cycle is annual, with purchases being made during the crop harvest season each year.
3) All straw in the area is subject to a unified purchase price and transportation unit price.
4) The collection distance is simplified by utilizing the collection radius within the studied area.
5) The transportation distance from the farmland to the biomass power plant is calculated by multiplying the straight-line distance by a tortuosity factor, without any roads occupying crop planting areas.

The crop resource quantity and straw collection parameters are detailed in Table 1. The methane yield per unit dry matter of straw is 350 m^3/t, with an average water content of the collected straw at 15 %. The straw collection radius is determined by the total amount of straw consumption:

$$R = \sqrt{\frac{M_{\mathrm{fuel}}}{\pi k \rho}} \tag{29}$$

The transportation and purchase cost of straw within a circular area are calculated using the following Eq. (30)- Eq. (31):

$$C_{\mathrm{T}} = \int_0^R \beta p_{\mathrm{t}} r \cdot \rho 2\pi r dr = \frac{2}{3}\pi\rho\beta p_{\mathrm{t}} R^3 \tag{30}$$

$$C_{\mathrm{B}} = M_{\mathrm{fuel}} \cdot p_{\mathrm{b0}} \tag{31}$$

Where, ρ represents the straw resource density, t/km^2, M_{fuel} is the total mass of straw consumed, t; k is a comprehensive coefficient determined by the straw collection coefficient and utilization coefficient, taking a value of 0.8; β is the distance tortuosity factor, taking a value of 1.5; p_{t} is the unit transportation price, 0.54$/(t•km); p_{b0} is the price per unit mass of straw, 32.9$/t; R is the collection radius, km.

Other costs, such as storage fees, loading and unloading fees, labor costs, etc., are directly proportional to the total collection cost, as following Eq. (32):

$$C_{\mathrm{o}} = \alpha S_{\mathrm{fuel}} \tag{32}$$

α represent the ratio of other costs to the total collection cost and has a value of 0.15.

Table 1
Crop resource quantity and straw collection parameters.

Crop types	Wheat	Rice	Soybean	Corn	Oil Bearing Crops	Tubers	Cotton	Sugar Crops
Regional crop yield	3.00	76.56	1.50	7.48	15.09	1.14	0.58	1.88
Grain-straw ratio	1.1	0.9	1.6	1.2	2.0	0.5	3.4	2.0
Crop collection coefficient	189.87	180.34	0.94	87.32	36.35	8.74	0.75	1.32

3.1.2. Environmental objectives

The study uses carbon emissions as an environmental indicator, consisting of three parts: emissions from system equipment operation, electricity purchase from the public grid, and biogas production. These are collectively referred to as *CE*.

$$CE = \sum_{i\in I}\sum_{i=1}^{8760} P_{i,t}^{out}\beta_i + \sum_{i=1}^{8760} P_{grid}^{out} em_g + \sum_{i=1}^{8760} V_{bio}^{in} em_{bio} \tag{33}$$

Where, *CE* represents the total carbon emissions over a period, kg; β_i is the carbon emission coefficient of equipment *i*, kg/kWh; em_g is the carbon emission factor of the regional grid, kg/kWh; em_{bio} is the carbon emission coefficient of biomass biogas.

3.1.3. Primary energy saving rate

The primary energy saving rate (PESR) is used in this study as an indicator to measure the improvement in energy efficiency in MECS. It is calculated by dividing the difference between the primary energy consumed by the tradition separate production (SP) system and that consumed by the MECS, by the primary energy consumed by the traditional SP system. Electricity consumption is converted into primary energy consumption using the following Eq. (34) [47,48].

$$PESR = 1 - \frac{PEC}{PEC_{con}} = 1 - \frac{\frac{E_{grid}}{\eta_{grid}\cdot\eta_e} + F_{ng}}{\frac{E_{gridcon}}{\eta_{grid,con}\cdot\eta_{e,con}} + F_{ng,con}} \tag{34}$$

Where, E_{grid} and $E_{gridcon}$ represent the electricity purchased from the grid by the MECS system and the traditional SP system respectively, kWh; F_{ng} and $F_{ng,con}$ represent the gas consumed by the MECS system and the traditional SP system respectively, kWh; η_{grid} and η_e represent the power generation efficiency of the power plant and the transmission efficiency of the grid, which are considered to be the same in both the MECS system and the SP system.

3.2. System optimization

3.2.1. Optimization model

The MECS planning considers three indicators: cost, environment, and primary energy saving rate. These aspects have conflicting characteristics and involve multiple decision variables. Thus, the complete planning model can be expressed as following Eq. (35) – Eq. (37):

$$\mathrm{optimal} f(X) = \mathrm{optimal}\,[f_1(X), f_2(X), f_3(X)] \tag{35}$$

$$X = \left[P_{WT,in}, P_{PGU,in}, P_{GB,in}, S_{PV,in}, S_{SC,in}, P_{CSHP,in}, P_{\frac{AC}{H},in}, P_{SS,in}, P_{TS,in}, n\right] \tag{36}$$

$$\text{Subject to}: X \in \Phi, P_{out}(X) = L(X) + \Delta(X), P_{out}(X) \in \Gamma \tag{37}$$

Where, *X* represents the decision variables, specifically the installed capacity of each equipment in this study; *f*(*X*) is the objective function representing the optimized performance of the MECS, with details introduced in Section 3.1; $X \in \Phi, P_{out}(X) = L(X) + \Delta(X), P_{out}(X) \in \Gamma$ are the constraint conditions, including equality constraints and inequality constraints, which will be introduced in the following sections.

3.2.1.1. Inequality constraint.

(1) Decision variable constraint

The selection of decision variables should be within a certain range:

$$P_i^{min} \le P_{i,in} \le P_i^{max} \tag{38}$$

$$S_{PV/SC}^{min} \le S_{PV/SC,in} \le S_{PV/SC}^{max} \tag{39}$$

Due to the limited space for installing PV and SC panels, the installation area of PV and SC is limited between *Smin PV/SC* and *Smin PV/SC*; The installed capacity of each equipment is also subject to certain limitations, with the installed capacity of the equipment of *i* not exceeding *Pmax i*, and not lower than *Pmin i*.

(2) Component operation restrictions

The operating power of each component of the system will not exceed its installed capacity:

$$0 \le P_i^{out} \le P_{i,in} \tag{40}$$

Furthermore, the output and remaining capacity of energy storage components are limited, with a restricted capacity for energy storage equipment.

$$SOC_k^L \le SOC_k(t) \le SOC_k^U \tag{41}$$

The limitations of the output are as follows:

$$\begin{aligned} 0 &\le P_{k,d}^t \le \gamma_{k,d}^{max} P_{k,in} \\ 0 &\le P_{k,ch}^t \le \gamma_{k,ch}^{max} P_{k,in} \end{aligned} \tag{42}$$

Where, *k* represent the type of energy storage equipment; *d* and *ch* represent the discharge and charge states of the energy storage equipment; *γmax k,d* and *γmax k,ch* represent the upper limits of single discharge and charge of the energy storage equipment; *SOCL k* and *SOCU k* represent the lower and upper bounds of the energy storage equipment state.

3.2.1.2. Equation constraint.

The equality constraints mainly involve the system's energy balance constraints:

$$P_{PV,e}^{Out,t} + P_{ICE,e}^{Out,t} + P_{WT,e}^{Out,t} + P_{PG,e}^{Out,t} + P_{BS,e}^{Out,t} = L_e^t + P_{GSHP,e}^{in} + \Delta L_e^t \tag{43}$$

$$Q_{SC,h}^{out,t} + Q_{TS,h}^{out,t} + Q_{AC,h}^{out,t} + Q_{GSHP,h}^{out,t} + Q_{EB,h}^{out,t} = L_h^t + L_{hw}^t + \Delta L_h^t + \Delta L_{hw}^t \tag{44}$$

$$Q_{AC,C}^{out,t} + Q_{GSHP,C}^{out,t} = L_c^t + \Delta L_c^t \tag{45}$$

3.2.2. Solution method

The solution process for multi-objective optimization models consists of two parts: finding the Pareto solution set, also known as the non-dominated solution set, in the first part; and selecting the ideal solution from this set as the desired solution in the second part, as shown in Fig. 4. The detailed process will be explained in the next section.

3.2.2.1. NSGA-II algorithm.

Inspired by biological evolution, the genetic algorithm, a global search intelligent algorithm, was proposed in 1975. It emulates natural selection and genetic mechanisms to systematically explore the solution space and identify the optimal solution. In 2002, Deb et al. [49] enhanced the NSGA algorithm by incorporating an elitist strategy and employing a fast non-dominated sorting method, there reducing its complexity and significantly improving its search capabilities. Currently, the NSGA-II algorithm has emerged as one of the

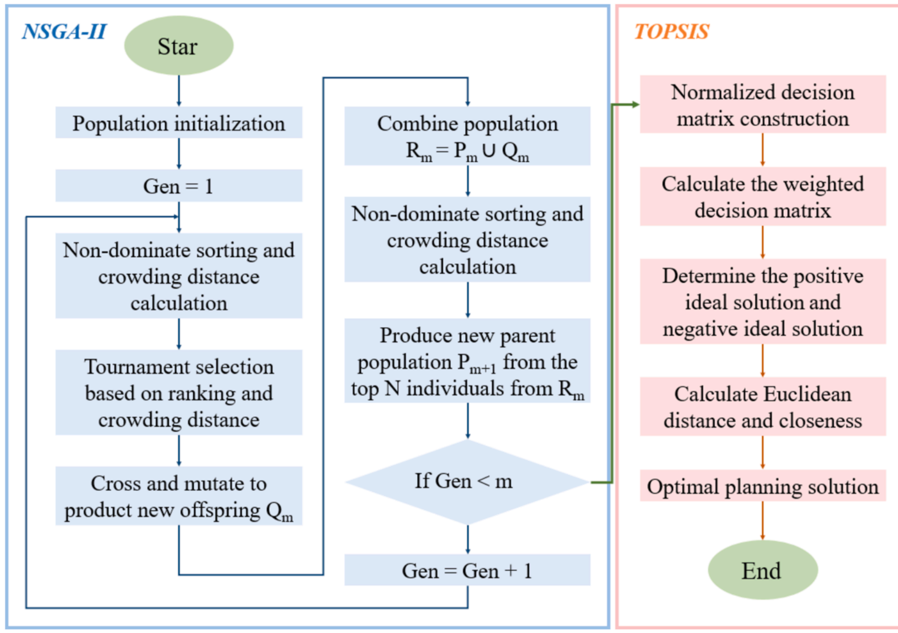


Fig. 4. Process of the solution method.

most extensively utilized approaches for addressing multi-objective optimization problems. Consequently, this study employs the NSGA-II algorithm as our chosen optimization technique.

3.2.2.2. TOPSIS method. The Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) is used to select the optimal solution, with the formula as following Eq. (46) and Eq. (47) [36]:

$$S_i^+ = \left[\sum_{j=1}^{m}\left(p_{ij} - \nu_j^+\right)^2\right]^{\frac{1}{2}} \tag{46}$$

$$S_i^- = \left[\sum_{j=1}^{m}\left(p_{ij} - \nu_j^-\right)^2\right]^{\frac{1}{2}} \tag{47}$$

Where, $\nu + j$represents the positive ideal solution for the objective j; ν- j represents the negative ideal solution for the objective j; $S + j$ is the distance from the target to the positive ideal solution $\nu + j$; S- j is the distance to the negative ideal solution ν- j.

$$C_i = \frac{S_i^-}{S_i^+ + S_i^-} \tag{48}$$

The higher the value of C_i, the better the target values become. The maximum value of C_i represent the optimal evaluation target.

3.3. Investigated case description

To assess the feasibility of implementing a MECS in rural areas, we selected a village in central China as a case study. Situated in Hubei province, this village benefits from abundant solar energy resources with long hours of sunlight throughout the year. This study used meteorological data and load data from rural areas of Wuhan, Hubei Province, China. The meteorological data came from NASA (https://asdc.larc.nasa.gov/project/SSE). Notably, sunlight intensity is highest at noon during summer and diminishes at night. The climate here is typically hot during summer and cold during winter, with varying wind velocities depending on the season. Additionally, the area boasts rich in biomass energy sources primarily derived from straw, an agricultural product suitable for biogas digestion facilities that yield sufficiently all year round. This study establishes three different modes.

Model 1: PG prioritizes supplying EV load, while BS does not charge.

Model 2: ICE and PG jointly handle EV load, while BS does not charge.

Model 3: ICE and PG jointly satisfy BS charging during valley periods and EV load.

The main input parameters for optimization the MECS capacity model include meteorological data from the study area, load demand simulations, MECS economic and technical parameters, and optimization algorithm settings. The regional load demand consists of EL, CL, HL and HWL required throughout the year, which are simulated in Dest as detailed in Fig. 5 and Fig. 6. The meteorological parameters of the study area include hourly temperature, solar irradiance and wind velocity for the entire year shown in Fig. 7. Note that cooling and heating seasons are determined based on building energy efficiency design standards while equipment usage is set according to actual survey findings. The MECS technical parameters consists of component-specific PLR models and ICE thermodynamic models listed in Table 2. The MECS economic parameter settings encompass initial investment costs and maintenance costs of each component detailed in Table 3.

Additionally, this study considers external costs of the MECS primarily originating from the power grid. The investigated case employs time-of-use pricing, with peak hours occurring from 11:00 AM to 1:00 PM and from 6:00 PM to 10:00 PM, at a grid price of 0.128 $/kWh;

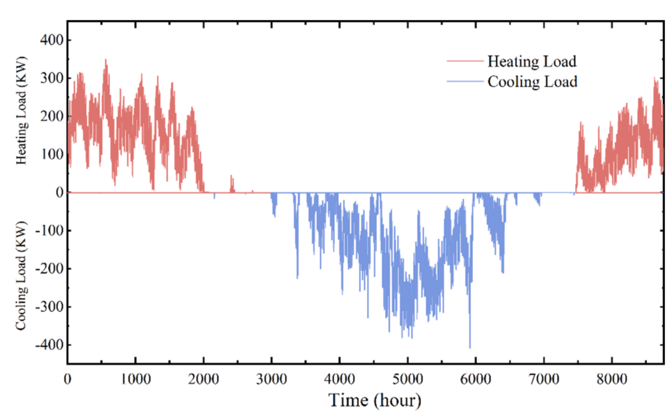


Fig. 5. Annual cooling and heating loads.

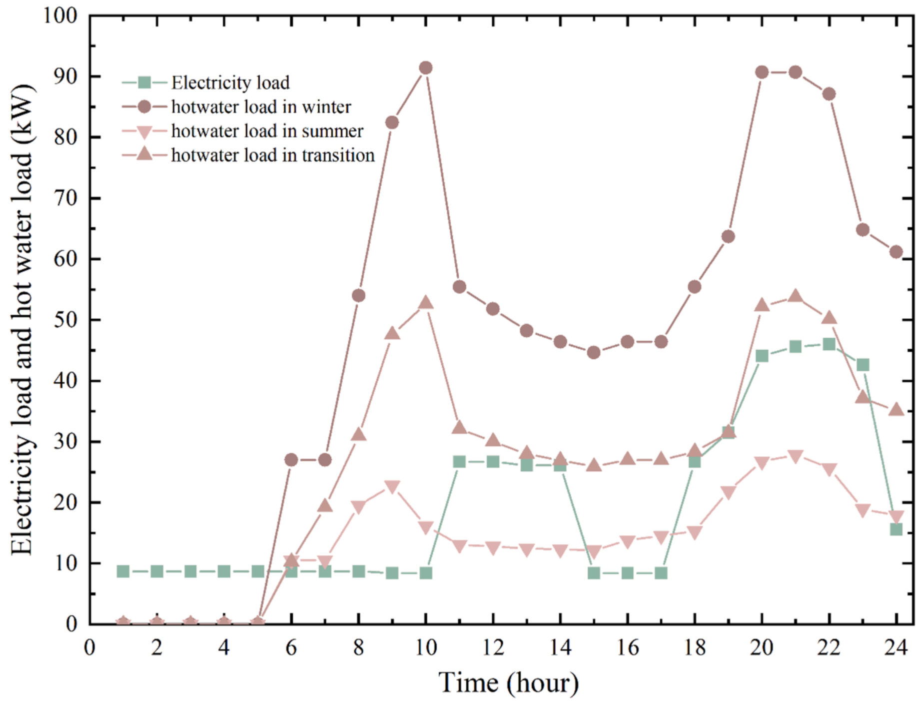


Fig. 6. Typical daily hot water and electrical load.

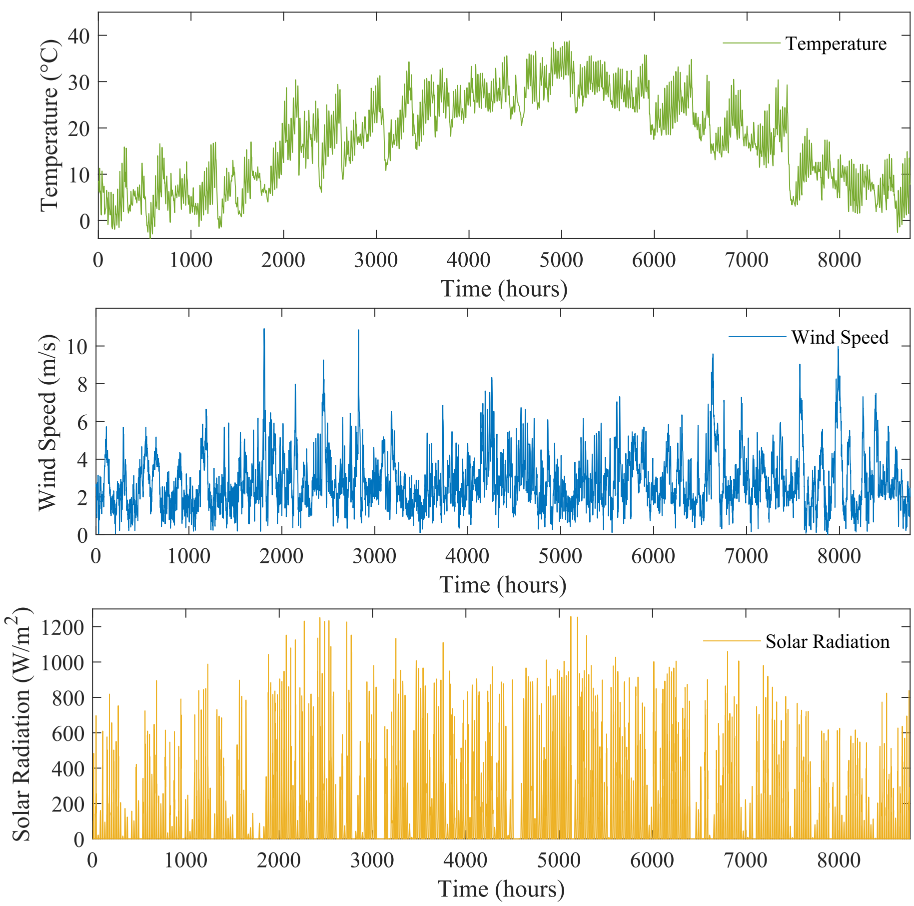


Fig. 7. Meteorological data used in the simulation study.

Table 2
Technical parameters of main devices.

Device	Parameters	Value
WT	$V_{\text{cut-in}}$	2.5
	$V_{\text{cut-out}}$	25
	V_r	9.0
SC	F_R	0.75
	$\eta_{SC,0}$	0.89
	α_0	5.40
	T_{in}	40
GSHP	$COP_{GSHP,ra}$(Cooling)	4.50
		5.00
	$COP_{GSHP,ra}$(Heating)	
AC/H	$COP_{AH,ra}$	1.40
BS	$\eta_{BS,chr}$	0.95
	$\eta_{BS,dis}$	0.90
TS	$\eta_{TS,chr}$	0.90
	$\eta_{TS,dis}$	0.90

Table 3
Economic parameters of main devices.

Device	Parameters	Unit	Value
PV [23]	Initial investment cost	$/m^2	364.5
	Maintenance cost	$/kWh	0.0015
	Service life	year	15
WT [50]	Initial investment cost	$/kW	600.0
	Maintenance cost	$/kWh	0.0015
	Service life	year	25
SC [23]	Initial investment cost	$/m^2	243.0
	Maintenance cost	$/kWh	0.0015
	Service life	year	30
ICE [18]	Initial investment cost	$/kW	734.9
	Maintenance cost	$/kWh	0.0006
	Service life	year	20
GSHP [23]	Initial investment cost	$/kW	436
	Maintenance cost	$/kWh	0.0014
	Service life	year	20
AC/H [18]	Initial investment cost	$/kW	225.0
	Maintenance cost	$/kWh	0.0004
	Service life	year	20
BS [23]	Initial investment cost	$/kWh	281.0
	Maintenance cost	$/kWh	0.0291
	Service life	year	13.5
TS [23]	Initial investment cost	$/kWh	14.0
	Maintenance cost	$/kWh	0.0063
	Service life	year	20
GS [51]	Initial investment cost	$/1	386
	Maintenance cost	$/year	58.1
	Service life	year	5

normal hours are observed between 7:00 AM and 10:00 AM, as well as between 2:00 PM to 5:00 PM, with a grid price of 0.078 $/kWh; and valley hours are from 0:00 AM to 6:00 AM and from 11:00 PM to 0:00 AM, with a grid price of 0.049 $/kWh. NSGA-II algorithm parameters including population size, maximum iteration count, and mutation crossover probability are outlined in Table 4.

Table 4
Parameters setting of the NSGA-II.

Parameters	Value
Population size	200
Genetic generations	1000
Crossover probability	0.9
Cross distribution coefficient	20
Mutual probability	0.1
Mutual distribution coefficient	20
Number of decision variables	9
Number of objectives	3

4. Results and discussion

The optimization results will be described and analyzed in detail, including the outcomes for different cost scenarios and the analysis of energy dispatch on typical days in each season. The simulations for this study were conducted using Matlab (2020b, the MathWorks, Inc, Massachusetts, USA) software.

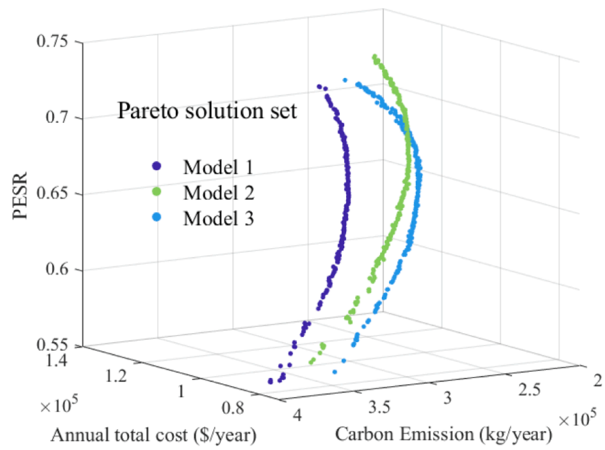
4.1. Optimizing pareto solution set analysis

As depicted in the figures, Fig. 8(a) represents the relative positional relationship of the Pareto solution set in space, while Fig. 8(b) illustrates the relationships among different metrics. The system undergoes multi-objective optimization based on three objective functions: economy, cost, and primary energy savings rate in three different modes. Each mode generates a Pareto frontier encompassing 200 MECS capacity planning scenarios, with each point in the figure represents a specific scenario. The Pareto boundaries of all scenarios are curved in three dimensions, indicating the trade-off relationship between the three optimization performances. Fig. 8(b) further demonstrates a negative correlation between cost and carbon emissions, as well as between carbon emissions and primary energy savings rate, while showing a positive correlation between annual cost and primary energy savings rate. The increase in investment cost primarily results from expanding renewable energy equipment capacity such like PV and WT. The utilization of solar and wind energy reduces the consumption of biomass energy and electricity purchased from the public grid, leading to an increase in primary energy savings rate. Furthermore, biogas engines and the public grid are major contributors to carbon emissions growth in the MECS. Therefore, increasing renewable energy capacity will correspondingly decrease carbon emission. Meanwhile, both biogas cogeneration and the grid primarily rely on combustion for power generation. While the grid consumes primary energy for generate electricity, excessive reliance on biogas cogeneration can also reduce energy efficiency due to incomplete combustion during power generation. Consequently, there is a negative correlation between carbon emissions and PESR.

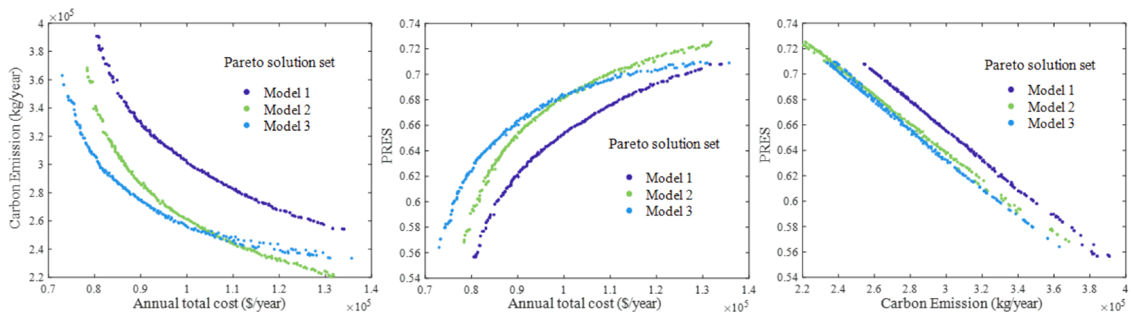
By comparing the Pareto solution sets of different modes in Fig. 8, it is observed that under the same total cost, Mode 1 has higher carbon emissions and lower primary energy savings rates than Mode 2 and Mode 3. This can be attributed to the fact that Mode 1 heavily relies on electricity purchases from the grid, with the batteries only performing passive energy storage rather than active storage. This finding suggests that even during periods of low electricity prices, procuring a substantial amount of electricity from the grid can still have adverse effects on the system's environmental performance and energy efficiency. Model 2 performs almost identically to Mode 3 in terms of primary energy savings rate. Although Mode 2 shows slightly better performance in carbon emissions and primary energy savings rate under high-scale costs, Mode 3 outperforms it in both aspects under low and medium-scale costs. This demonstrates that utilizing battery storage (BS) during periods of low demand can effectively improve the economic performance and energy efficiency of the system. The objectives of Integrated Energy System (IES) capacity planning are to minimize the annual total cost, reduce the annual total carbon emissions, and maximize the primary energy savings rate. Therefore, selecting an ideal solution from the Pareto optimal solution set is necessary for simultaneous optimization of cost, environment and energy efficiency. The next section will introduce the MECS planning capacity selected through the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) method.

4.2. Optimal decision variables and optimization performance analysis

The optimized capacities of various equipment under different modes for the MECS are displayed in Table 5. The optimized annualized investment costs, electricity purchase costs, operating costs, straw



(a) Pareto solution set in three dimensions



(b) The relationships between the three indexes

Fig. 8. Pareto-optimal solution set.

Table 5
Configuration planning of various devices in different scenarios.

Scheme	PV(m^3)	SC(m^3)	WT(kW)	ICE(kW)	GSHP(kW)	AC/H(kW)	BS(kWh)	TS(kWh)	GS(n)
Model 1	455	529	98	51	176	64	131	338	81
Model 2	401	510	100	52	183	69	79	331	82
Model 3	199	539	89	62	185	69	83	338	88

Table 6
Planning target results in different scenarios.

Scheme	Model 1	Model 2	Model 3
Annualized investment cost(k$/year)	72.79	70.91	62.65
Annual electricity purchase cost(k$/year)	12.88	8.25	8.01
Annual operation and maintenance costs(k $/year)	7.97	7.62	7.65
Annual collection cost of straw raw materials(k$/year)	12.85	16.50	17.09
Total cost(k$/year)	106.49	103.29	95.41
Total carbon emissions(t/year)	288.50	254.67	262.30
Primary energy saving rate	66.89 %	68.93 %	67.52 %

collection costs, total costs, annual total carbon emissions, and primary energy savings rates under different modes for the MECS are presented in Table 6 and Fig. 9. As can be seen from Table 5, on the supply side, the changes in planned capacity mainly concentrate on PV and BS equipment with little variation in wind energy equipment's planned capacity. For different modes, the optimization results of Mode 1 yielded a relatively high level of redundant configuration, with the highest PV and BS capacities among the three modes, reaching 455 m^3 and 131 kWh respectively. In particular, the PV capacity increased by 11.87 % and 76.04 % compared to Mode 2 and Mode 3, respectively, while the BS capacity increased by 39.69 % and 36.64 %, respectively. Consequently, the investment cost for Mode 1 is the highest among the three modes, with an annualized investment cost reaching 72.79 k$/year. Using only PG to meet the power supply demand of EVs also resulted in the highest electricity purchase cost among the three modes, reaching 12.88 k $/year. Therefore, even though the total cost of straw collection in Mode 1 decreased by 28.40 % and 32.99 % compared to Mode 2 and Mode 3, respectively, its overall cost is still higher than that of Mode 2 and Mode 3. The system cost of energy stands at 0.19 $/kWh, which is also the highest among the three modes. So, even though the total cost of straw collection in Mode 1 decreased by 28.40 % and 32.99 % compared to Mode 2 and Mode 3, respectively, its overall cost remains higher than that of Mode 2 and Mode 3. Moreover, extensive purchases of electricity from PG also decrease the primary energy savings rate of Mode 1. The optimization results of Mode 2 indicate that the supply-side equipment

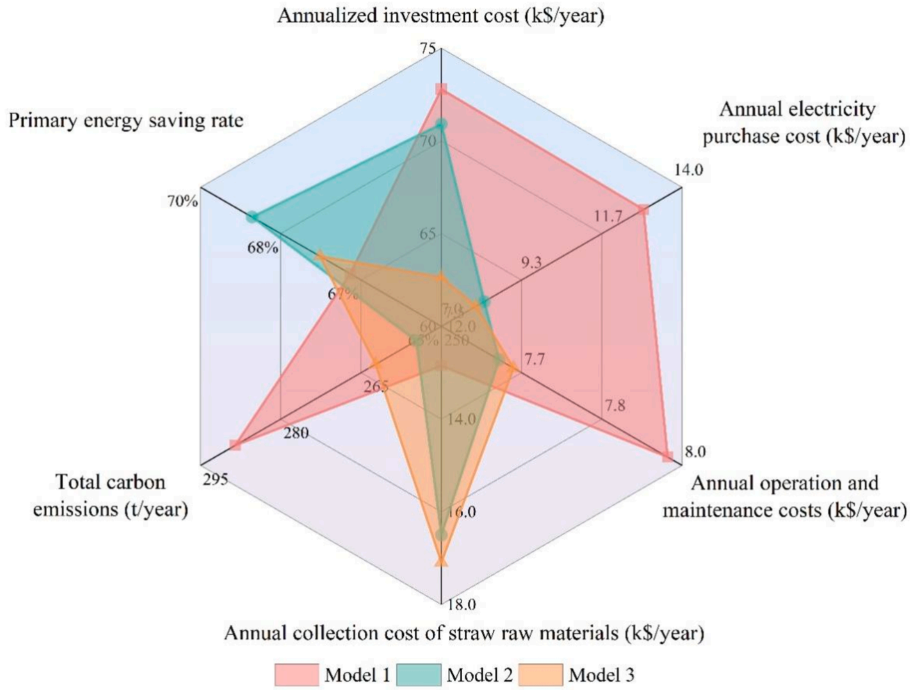


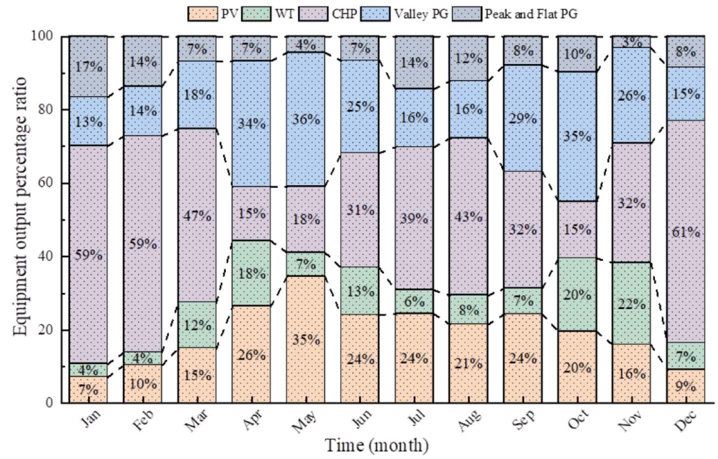
Fig. 9. Comparison of optimization results in different modes.

redundancy configuration after optimization is significantly smaller than that of Mode 1, with the planned capacities of PV and BS reduced to 401 m^3 and 79 kWh, respectively. Nevertheless, its redundancy configuration remains higher than that of Mode 3. Therefore, the investment cost of Mode 2 is lower than that of Mode 1 but higher than that of Mode 3. This results in Mode 2 having a lower investment cost than Mode 1 but a higher cost than Mode 3. Additionally, its electricity purchasing cost has significantly decreased to 8.25 k$/year, a reduction of 35.94 % and the overall cost of energy has dropped to 0.18 $/kWh. Mode 3′s PV capacity planning is significantly lower than that of Mode 1 and Mode 2, at only 199 m^3, while the ICE capacity is slightly higher than Mode 1 and Mode 2, with increases of 17.74 % and 16.12 %, respectively. This is because Mode 3 utilizes valley periods for energy storage, greatly enhancing the regulation capability of BS in the system. In contrast, in Modes 1 and 2, BS can only store excess electricity generated by PV and WT, providing weaker regulation capabilities. Therefore, redundant capacity should be planned to meet electrical loads. As shown in Table 6, even though Mode 3 needs to purchase a portion of electricity from the grid during the night to meet the charging needs of BS, its total electricity purchasing cost is still the lowest among the three modes, at only 8.01 k$/year. Although the total cost of straw collection in Mode 3 is slightly higher than that in Mode 2, overall, Mode 3 can generate favorable economic effects. Compared to Mode 1 and Mode 2, its total cost decreases by 11.61 % and 8.26 %, respectively. The cost of energy has decreased to 0.15 $/kWh. Li et al.[31] conducted a comprehensive and in-depth analysis of an off-grid hybrid solar-wind-biomass power system, fully demonstrating the feasibility of this system for electrification in remote rural areas. Their work laid a very solid foundation for subsequent research and the system energy cost calculated by the research institute was 0.201 $/kWh, slightly higher than that of this study. The distinguishing feature of this study is the inclusion of efficient cooling and heating equipment in the system model. By incorporating real-time partial load ratios and cogeneration modes, further reductions in energy costs can be achieved, as also reflected in literature [20]. In terms of the environment, as ICE undertakes more loads in the system, the carbon emissions of Mode 3 reach 262.30 t/year, an increase of 7.63 tons compared to Mode 2 but a decrease of 26.2 tons compared to Mode 1.

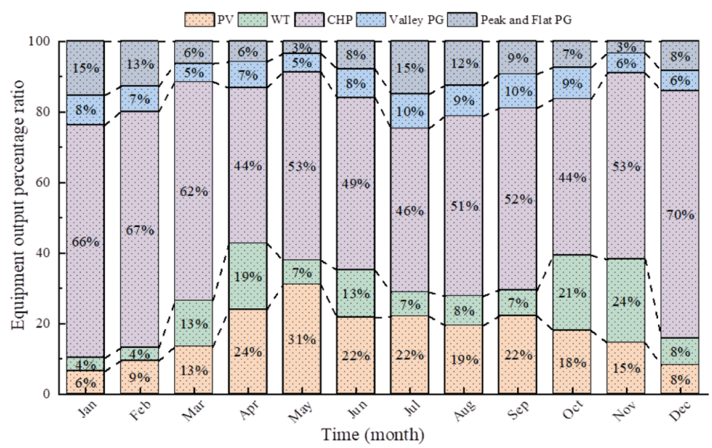
The monthly output ratios of the devices in the multi-energy complementary system under different modes are illustrated in Fig. 10. It can be observed from Fig. 10 that in Model 1, the output proportions of PV and WT exceed 40 % in three months and are higher than 30 % in a total of seven months. The average output proportion is slightly higher than that of Mode 2 and significantly higher than that of Mode 3. In Modes 2 and 3, biogas cogeneration dominates the system’s output. The use of ICE at night to meet the power demand of EVs also significantly reduces the power purchased from the grid. The average proportion of total power purchased from the grid decreases from 30.67 % in Mode 1 to 16.25 % in Mode 2 and 16.08 % in Mode 3. In Mode 2, the average proportion of power purchased during off-peak hours is 7.5 %, and the average proportion of power purchased during non-off-peak hours is 8.75 %. In Mode 3, a higher proportion of power is purchased during off-peak hours, reaching an average of 9.83 %, while a lower proportion of power is purchased during non-off-peak hours, with an average of only 6.25 %. This indicates that Mode 3 makes more efficient use of off-peak power than Mode 2, playing a role in peak shaving and valley filling, and better mitigating system fluctuations across different time periods. For different months of the year, regardless of the mode, PV and WT output are the highest in April, May, June, October and November. There are two main reasons for this. Firstly, solar radiation intensity and wind speed are relatively higher in these months compared to winter resulting in greater overall output from PV and WT. Secondly, cooling or heating load is almost zero during these periods with significantly lower overall electricity demand compared to summer and winter. Therefore, even though solar radiation intensity is higher in summer allowing PV to provide more electricity supply, ICE still accounts for a larger proportion of energy supply.

4.3. Analysis of system operating characteristics

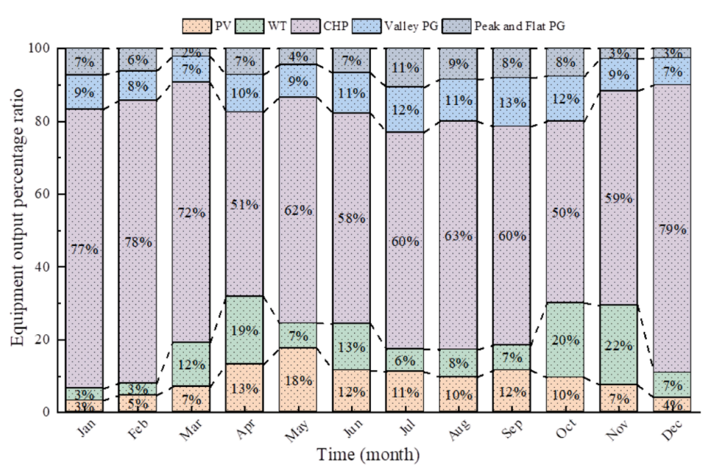
The typical daily electrical load charts of the integrated energy system in summer, winter, and transition seasons are presented in Fig. 11 (a)-(i). In this study, we employed a method for selecting typical days by collecting hourly data from various seasons throughout the year and utilizing the k-Means clustering analysis algorithm [52,53]. Specifically, February 18th was identified as a representative day for winter, August



(a) Monthly proportion of equipment output in Model 1



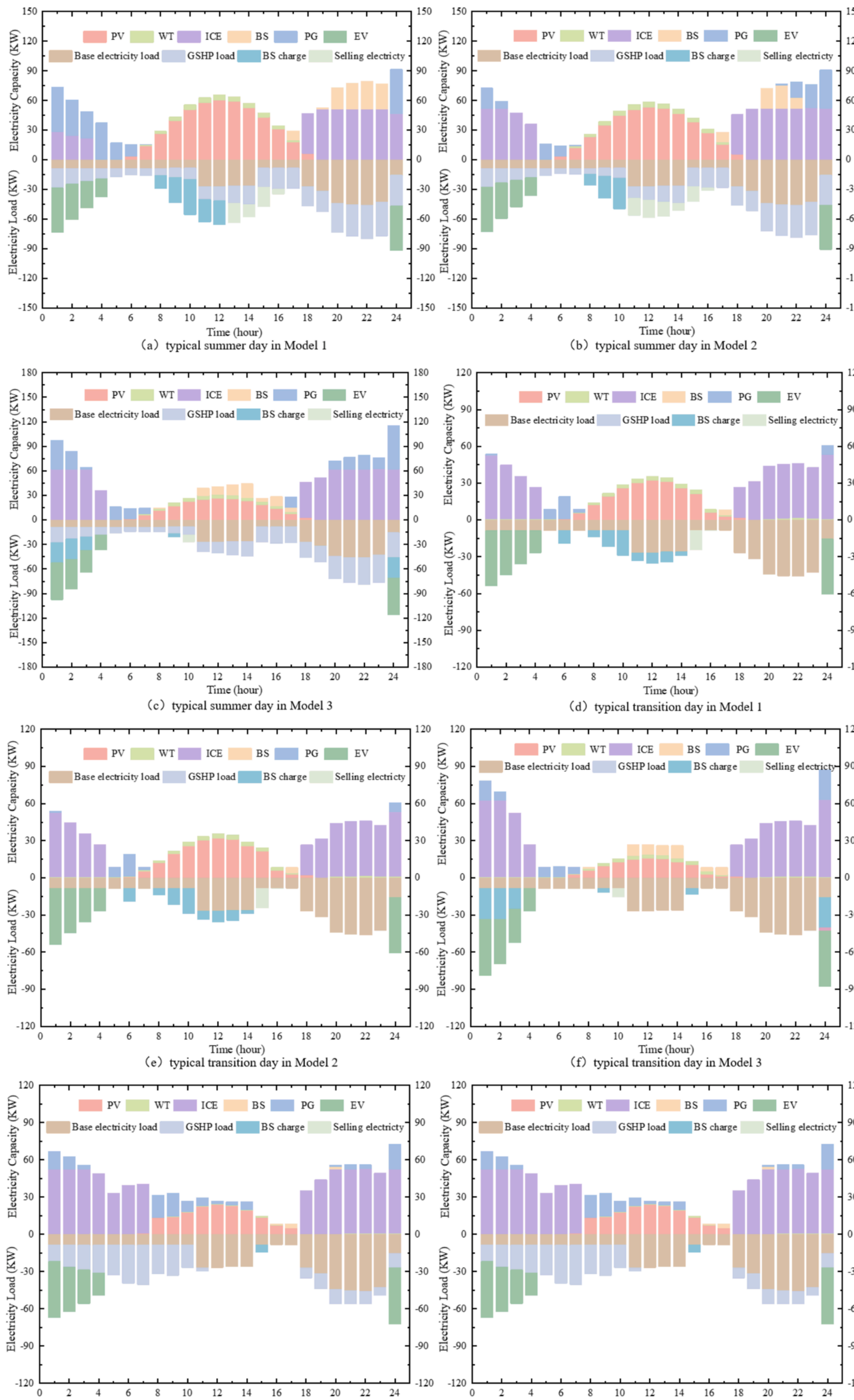
(b) Monthly proportion of equipment output in Model 2



(c) Monthly proportion of equipment output in Model 3

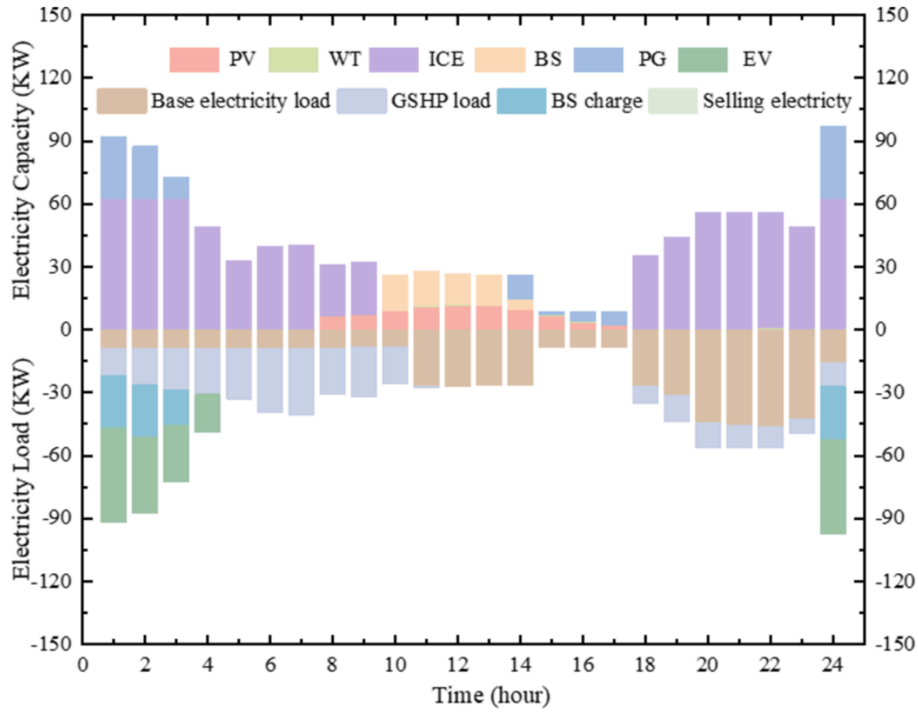
Fig. 10. Monthly output ratio of each equipment.

15th for summer, and April 22nd for the transition season. During the nighttime in transition seasons, since there is almost no cooling or heating load, the electrical load consists only of basic electrical load and EV load. In Mode 1, as the electricity required by EVs is directly provided by PG, the remaining electrical load is much lower than the minimum startup load of ICE. Therefore, at this time, PG completely meets the electricity demand. In Mode 2 and Mode 3, most of the required electricity is supplied by ICE. The situations during winter and summer



(a) typical summer day in Model 1
(b) typical summer day in Model 2
(c) typical summer day in Model 3
(d) typical transition day in Model 1
(e) typical transition day in Model 2
(f) typical transition day in Model 3
(g) typical winter day in Model 1
(h) typical winter day in Model 2

Fig. 11. Hourly electricity load in a typical day.



(i) typical winter day in Model 3

Fig. 11. (*continued*).

nights are similar to that in transition seasons; however, in Mode 3, starting from 11:00 PM onwards, which corresponds to the valley electricity pricing period, the system adopts a cyclic charging strategy for battery storage. As a result, the overall load during this period is slightly higher compared to Mode 1 and Mode 2. Additionally, because Mode 3 plans for a larger ICE capacity compared to Mode 1 and Mode 2, the minimum startup value of ICE in Mode 3 is higher as well. During the time period from 5:00 to 7:00 in the morning, users' electricity demand is less than the minimum startup value of ICE,and PG still needs to provide power.

In summer and transitional seasons, the photovoltaic system usually starts generating electricity at 7:00 AM, and in winter, it typically starts one hour later. Between 7:00 AM and 10:00 AM, which corresponds to the period of normal electricity pricing, Mode 1 and Mode 2 exhibit a significant discrepancy between the power output of the equipment and the load curve due to excessively PV over-allocation. Consequently, reliance on BS becomes necessary to store surplus electricity. In contrast, Mode 3′s planned equipment demonstrates an almost perfect alignment between its power output curve and the load curve during this time frame, indicating a more reasonable planning approach. As solar radiation intensity increases, PV power generation gradually rises and reaches its peak between 11:00 AM and 1:00PM, followed by a subsequent decline. During this period, if the BS operates at maximum capacity in Mode 1 and Mode 2, surplus power needs to be transmitted to the grid for full consumption. However, in Mode 3, no excess power is generated; instead, the system supplements electricity through the BS to adequately meet the user's load demand. In winter, regardless of the mode, when the PV output reaches a certain value, the remaining electrical load does not meet the startup conditions of ICE. At this time, other power facilities need to be used for supplementation. Mode 1 and Mode 2 can only use PG for supplementation since there is no excess electricity stored in BS. However, In Mode 3, BS provides a significant portion of the deficient electricity. Specifically, it averages 50.15 % in winter, 32.86 % in spring, and 28.56 % in summer. This substantially reduces the electricity purchased from PG and alleviates the pressure that the system places on the public grid. After 6:00p.m., both photovoltaic and wind power generation decrease close to zero while user's electrical load demand begins to increase and reaches its peak between 8:00p.m. and 11:00p.m.. During this period, ICE becomes the main equipment providing required electricity for users; however, it cannot fully meet their demand at peak times so some electricity still needs to be provided by PG although overall it is not significant.

The typical daily hot water load in summer under different modes is shown in Fig. 12(a)-(c), while the load in transition seasons is depicted in Fig. 12(d)-(f). Additionally, the typical daily hot water load in winter can be found in Fig. 12(g)-(i). During winter nights, the hot water load is zero, while the heat load gradually increases. Under all three modes, AH and GSHP jointly provide the heat demand of users, with AH mainly driven by waste heat generated by ICE. SC starts supplying the required heat to users at 8:00 a.m., gradually increases as solar radiation intensity increases. Between 8:00 a.m. and 10:00 a.m., the sum of heat load and hot water load reaches its peak for the day. During this time, under various modes, SC, AH, and GSHP coordinate to provide users' demands for hot water hand heating. After 10:00 a.m., as users' heat demand of significantly decreases, SC can meet most of it; however, during evenings when there is an increase in hot water load, both AH and GSHP are still needed to provide heating and hot water loads respectively, with AH still primarily driven by waste heat generated by ICE. Additionally, TS stores little heat in winter, so its heat storage function is limited. In summer and transition seasons, there is only a hot water load and no heating load. Compared to winter, the stronger solar radiation and lower hot water load make SC sufficient to meet all the hot water demands of users during the day, with a large amount of excess heat stored in TS for use at night. The typical daily cooling load in summer is shown in Fig. 13. The cooling demand in summer is mainly provided by AC and GSHP. AC operates at full capacity while GSHP is used to meet the remaining cooling demand of users. From 5:00 a.m. to 5:00p.m., as PV provides most of the power demand for users, ICE remains turned off and does not generate waste heat. During this time, AC directly uses the heat generated by biogas combustion. In other time periods, the heat provided to AC mainly comes from the waste heat generated by ICE.

5. Conclusion

To fully utilize rural spatial resources such as rooftops and renewable energy sources like wind, solar, biomass, and geothermal energy, as well as optimize the utilization of valley electricity under the time-of-use electricity pricing mechanism, this study proposes a collaborative planning method for rural a multi-energy complementary system (MECS). This method encompasses various power supply, cooling and

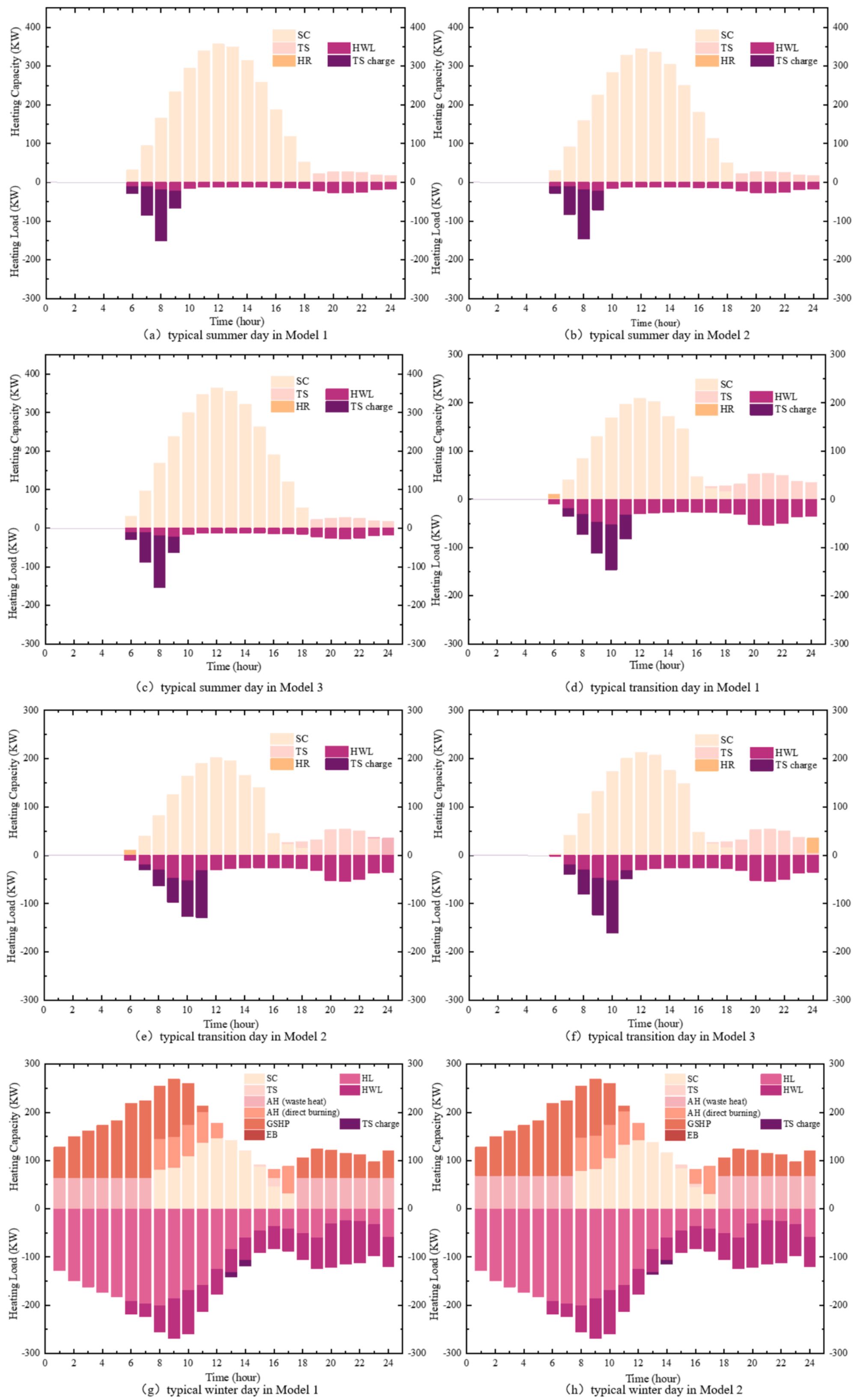
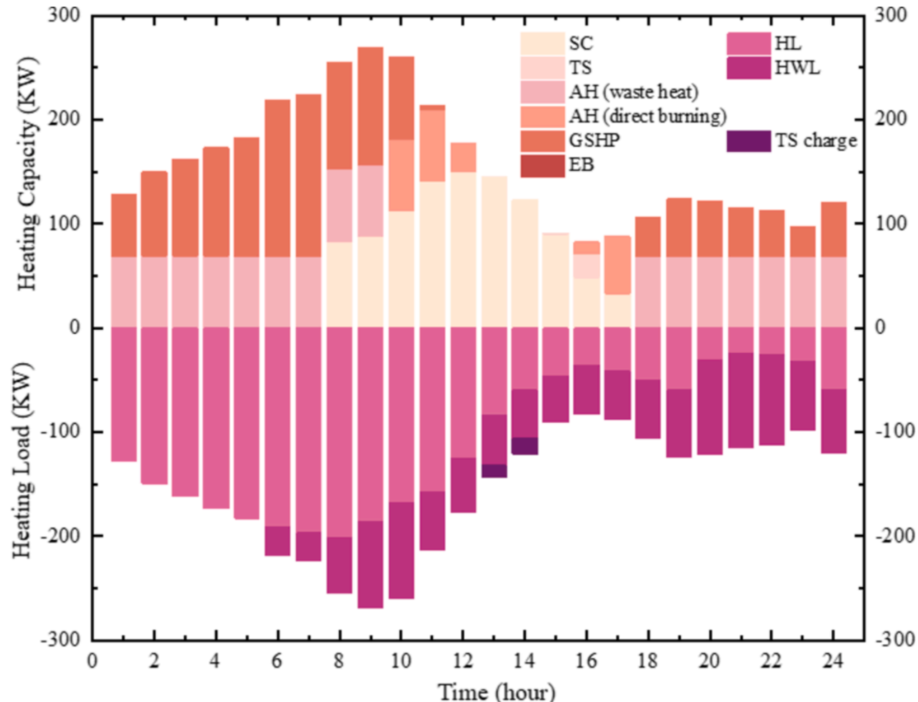
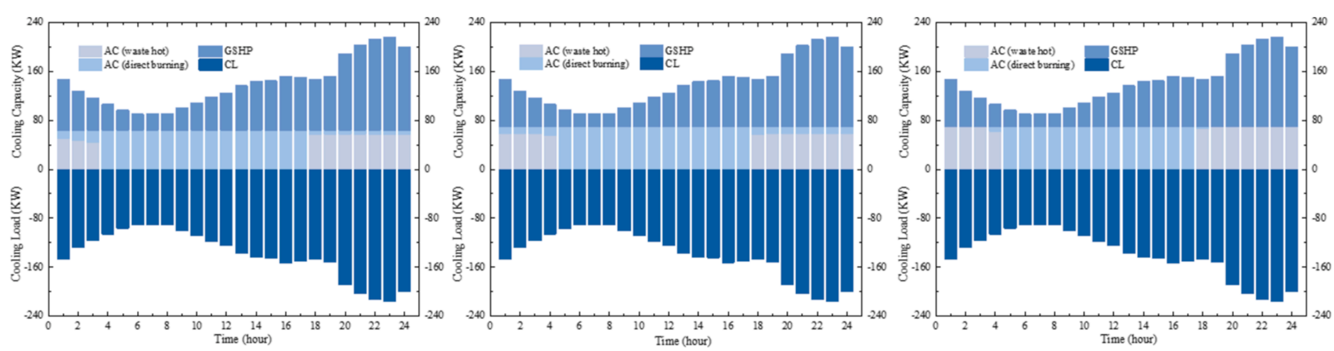


Fig. 12. Hourly hot water and heating loads in a typical day.



(i) typical winter day in Model 3

Fig. 12. (*continued*).



(a) typical summer day in Model 1 (b) typical summer day in Model 2 (c) typical summer day in Model 3

Fig. 13. Hourly cooling load in a summer typical day.

heating equipment, and establishes a multi-objective optimization model aiming to minimize the annual total cost and carbon emissions while maximizing primary energy savings rate. Taking a rural area in central China as a case study leads to the following conclusions:

1) In the multi-energy complementary system, the biomass gas-fired combined heat and power (CHP) unit serves as the primary energy supply equipment, while photovoltaic (PV) and wind turbines (WT) act as auxiliary energy supply equipment. Considering the partial load ratio (PLR) of each equipment, especially in relation to the start-stop operation of the internal combustion engines (ICE), coupling between PV, WT, and ICE becomes more crucial. Battery storage (BS) effectively enhances their coupling capabilities and significantly reducing grid power purchases.
2) The surplus power generated by PV and WT is the main source of electricity stored in BS in Mode 1 and Mode 2, resulting in redundant planning capacity for PV and WT. However, when BS actively stores electricity during the valley period of the electricity price, it enhances the complementary capabilities among BS, PV and WT during system operation. This reduces the planning capacity of PV and WT in the MECS while increasing ICE planning capacity, enhancing the system's ability to synergistically provide electricity and heat. As a result, the system investment cost has been reduced by 16.19 % and 13.18 %, respectively.
3) Purchasing large amounts of electricity from the grid during the vally period of electricity prices has negative impacts on the system's economy, environment, and primary energy savings rate. Utilizing more biomass gas-fired CHP can enhance the economic, environmental, and primary energy savings benefits of the system.

In summary, the MECS can meet rural areas' cooling, heating, and electricity demands by utilizing various forms of complementary energy. However, future research should address issues such as considering the elasticity of raw material supply prices due to abundant biomass production materials and the impact of raw material collection quantities in different regions. Additionally, further research is needed on the impact of BS and EV scheduling strategies on system reliability and complementarity.

CRediT authorship contribution statement

Min Chen: Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Investigation, Conceptualization. **Jiayuan Wei:** Writing – original draft, Validation, Investigation. **Xianting Yang:** Investigation. **Qiang Fu:** Investigation. **Qingyu Wang:** Investigation. **Sijia Qiao:** Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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